

## Education

### University of Salento

PH.D IN THEORETICAL PHYSICS

Feb 2023 - Feb 2026 (Ongoing) Lecce, Italy

- 11 peer-reviewed publications in leading journals; full list available on **InspireHEP**
- Presented research at international conferences: 3 talks and 1 poster.
- Mentored a fellow PhD student, introducing them to the group's research and methods.
- Attended 11 international workshops and schools on advanced topics in quantum field theory, quantum gravity, and computational physics.

POST GRADUATE MASTER'S DEGREE IN APPLIED ARTIFICIAL INTELLIGENCE - PYTHON, PYTORCH, NOSQL, SCIKIT-LEARN

Jan 2025 - Jan 2026 (Ongoing) Lecce, Italy

- Focused on machine learning, deep learning, data analysis, and big data.

### ISUFI (University Institute of Interdisciplinary Education)

POSTGRADUATE MASTER'S DEGREE - FINAL GRADE: 70/70 WITH HONORS

Oct 2018 - May 2023 Lecce, Italy

- Honors program for top students, offering interdisciplinary exams, language training, and a merit-based scholarship including accommodation.

### University of Salento

M.S. IN THEORETICAL PHYSICS - FINAL GRADE: 110/110 WITH HONORS

Apr 2021 - Sep 2022 Lecce, Italy

- Recipient of a competitive merit-based scholarship covering tuition, accommodation, and a monthly stipend, renewed for the entire M.S. program.

B.S. IN PHYSICS - FINAL GRADE: 110/110 WITH HONORS

Oct 2017 - Feb 2021 Lecce, Italy

- Received a top 1% merit scholarship covering full tuition, housing, and a monthly stipend for the entire B.S. program.
- Elected Student Representative on the Department and Teaching Council for a 2-year term.

## Work Experience

### GraphAware

AI ENGINEER INTERN - PYTORCH, MLX-LM, UNSLOTH TRANSFORMERS, NEO4J

Sep 2025 - Dec 2025 (Ongoing) Lecce, Italy

- Fine-tuning GPT-oss-20b to convert graph database queries into html human-readable reports, using also GRPO to improve reasoning quality.
- Building a reinforcement-learning pipeline that fine-tunes a small LLM to generate personalized content using engagement metrics (likes, impressions, feedback) as rewards to adapt and optimize content quality and post relevance automating the content-creation workflow.

### Federico II University of Naples, Department of Agricultural Sciences

PROFESSIONAL CONSULTANT (FREELANCE CONTRACT) - PYTHON, HARDWARE

May 2025 - Jul 2025 Naples, Italy

- Designed and accomplished a measurement system to monitor CO<sub>2</sub> and O<sub>2</sub> production in biological systems, enabling real-time assessment of environmental changes.
- Optimized system architecture, reducing costs by 30% while maintaining accuracy and reliability of data acquisition.
- Developed the Python pipeline for sensor integration, real-time data processing, and visualization in a custom desktop app, supporting input from 26 sensors and improving research workflow efficiency and usability.

## Personal Project

### H2Algae - Monitoring System for Hydrogen Production from Microalgae - Team Project

PYTHON, HARDWARE

PoC, Git

- Designed and optimized grow and sensor chambers, integrating hardware with Python-based data acquisition.
- Won the Makerspace Call, participated in entrepreneurial programs PDAI 2023 and Call for Startups 2023 and presented at Startup Day 2023.
- Led to the consulting engagement with the University of Naples, validating the systems for real-time gas monitoring in biological research.

### Deep Learning for Geospatial Temperature Season Classification and Forecasting

PYTORCH, TRANSFORMERS, CNN

Git

- Built a CNN classifier for temporal attribution, converting weather maps into a labeled month-prediction dataset reaching an accuracy of 99.32%.
- Developed U-Net-based forecasting models using NetCDF temperature data improving by 22% the 9 hours predictions compared to the baseline.
- Testing Vision Transformers to solve the same weather-map prediction problem improving by 27% the 21 hours prediction with respect to the baseline.

## Skills

|                          |                                                              |
|--------------------------|--------------------------------------------------------------|
| <b>Languages</b>         | Python, C/C++, SQL, LaTeX                                    |
| <b>ML/AI</b>             | PyTorch, Mlx-lm, Unsloth, Ollama, Hugging Face, scikit-learn |
| <b>Data &amp; Graph</b>  | Neo4j, Redis, ClickHouse, Pandas, Dask                       |
| <b>Software</b>          | Git, Docker, Mathematica                                     |
| <b>Foundations</b>       | Algorithms, data structures, numerical methods               |
| <b>Languages (human)</b> | Italian (native), English (C1), French (B1)                  |